§ . . Anchor Draft — Token-efficiency mandate

Revision: 1 **Last modified:** 2026-06-09T08:30:00Z **Status:** draft (for conductor to land in `Constitution.md` + mirror into CLAUDE.md / AGENTS.md / QWEN.md) **Anchor slot:** §11.4.141 (verified free 2026-06-09; §11.4.140 reserved by `action_prefix_system`).

> Two blocks below: (A) the canonical anchor for `constitution/Constitution.md`, and (B) the consumer-side mirror block for every consumer CLAUDE.md / AGENTS.md / QWEN.md (identical content discipline per §11.4.35; the consumer block adds the project instantiation pointer).

(A) Canonical anchor — for `constitution/Constitution.md`

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§ . . — Token-efficiency mandate (research-derived + operator mandate, - -)

Forensic anchor (operator mandate, 2026-06-09): the AI coding agent's token spend (input AND output) is dominated by an enormous always-loaded governance context — the consumer CLAUDE.md/AGENTS.md/QWEN.md plus the constitution submodule (measured ~170,000 tokens of static governance re-sent on EVERY turn: consumer `CLAUDE.md` ~90.5K + `constitution/CLAUDE.md` ~79.6K, a 112-row Applied-Fixes table plus verbatim §11.4.X anchor restatements) — compounded by 350–520K-token subagent transcripts, verbose status, full-file re-reads, and full-file loads where retrieval would do. The same pattern is externally

documented as a scaling defect (Claude Code issue #24147 — "Cache read tokens consume 99.93% of usage quota - architectural scaling issue with CLAUDE.md re-reads"). Operator mandate: cut token spend to **30–40% of current (a 60–70% reduction)** WITHOUT degrading quality, performance, or safety, and WITHOUT breaking any existing mechanism.

Every project worked on by AI coding agents MUST adopt a token-efficiency regime composed of the following measures, ranked by safety and impact, and MUST PROVE the reduction with a measured anti-bluff harness — never an asserted number (§11.4.6 / §11.4.123). The headline 60–70% is the warm-cache target; the rule requires the *measured* best-safe reduction with cited evidence, never the estimate.

The measures (composable, each preserving every existing rule):

1. **Prompt-cache the static governance prefix (PRIMARY — the single biggest lever).** The always-loaded governance MUST form a byte-stable prefix with a cache breakpoint at its end, with nothing volatile rendered ahead of it. Cache reads cost ~0.1× base input price (a 90% discount); writes 1.25× (5-min TTL) / 2× (1-hour). The discount applies to the dominant cost component. **Caching is transparent — the model sees byte-identical input cached or not — so it CANNOT remove a rule, weaken a gate, lower quality, or change a verdict; it changes only billing.** The only failure mode is a *silent invalidator* (timestamp/UUID/unsorted-JSON ahead of the breakpoint), which costs money but never corrupts, and is detectable via `cache_read_input_tokens`. Claude Code caches natively; SDK callers use `cache_control:{type:"ephemeral"}`; Gemini/Qwen use their context cache via API-key auth.
2. **Subagent model-tiering + output-to-file (SECONDARY — biggest non-cache win).** Route MECHANICAL, NON-JUDGMENT subagent work (search, grep, status, doc-export, file-presence, read-only probes) to a cheaper/smaller model (Haiku-class); RESERVE the strong model for ALL reasoning, fix-design (§11.4.102), verdicts, code-review (§11.4.125), and demotion-evidence (§11.4.7). Subagents persist large output to a file and return a pointer instead of a 350–520K-token inline transcript. **The strong model owns every decision and every PASS/FAIL — the cheap model never emits a verdict — so**

§11.4.50 determinism and the anti-bluff covenant are untouched; quality cannot degrade. Bounded by §12.6 (60% memory) and the §11.4.58/§11.4.103 stream caps.

3. **Thin always-loaded INDEX + on-demand detail (progressive disclosure).** Restructure the consumer governance so the always-loaded file is a concise index (one line per fix / per anchor) and the full bodies are fetched on demand. **Three invariants keep every gate green and every rule reachable: (a) each index line carries the literal `11.4.N` anchor token so the `CM-COVENANT-114-N-PROPAGATION` literal-anchor scans still pass; (b) the canonical full text stays in the tracked, gate-scanned `constitution/Constitution.md` — no rule is deleted, only de-duplicated out of the consumer; (c) the full body is reachable in one hop.** This realises §11.4.35's consumer-thin / submodule-canonical split at the byte level — a de-duplication, never a deletion.
4. **Retrieval / CodeGraph-first over full-file loading.** Structural questions (where is X / what calls Y / what would break) → CodeGraph (§11.4.78/§11.4.79, already mandated + installed) or semantic search; region questions → grep-scoped read; never re-read a harness-tracked file. Read-only, already-trusted (§11.4.78) — no behaviour change.
5. **Output-token reduction.** Terse conductor status (no restatement of unchanged governance); `effort:"low"` on the mechanical-subagent allowlist only; structured output where a sink consumes it. Output is ~5× input price, so this is cost-leveraged. Presentation-only — captured-evidence artefacts (§11.4.5/§11.4.69) unchanged.
6. **Tool-call efficiency.** Batch independent tool calls in one block; reuse harness-tracked file state; reuse persisted tool outputs.
7. **Compaction / context-editing for long sessions (defensive).** Prune stale tool-results / old thinking so a 100+ turn cycle doesn't re-pay for an ever-growing transcript; never touches the governance prefix or on-disk evidence.

Mandatory measured proof (anti-bluff — §11.4.6/§11.4.50/§11.4.69/§11.4.123): the reduction MUST be proven by a token-accounting harness measuring tokens-per-development-cycle BEFORE vs AFTER on a frozen deterministic workload, split into `input_tokens` / `cache_read_input_tokens` / `cache_creation_input_tokens` / `output_tokens` from the authoritative

Anthropic `usage` object (NEVER `tiktoken`; NEVER the client-side `total_cost_usd` estimate — recompute cost from raw token fields with the published prices), reproduced N times to identical verdict (§11.4.50). Pass = AFTER ≤ 40% of BEFORE (target) OR the measured best-safe reduction with a cited cold-cache reason. The AFTER run MUST ALSO show ZERO regression on the pre-build full sweep, the meta-test mutation sweep (every gate still FAILs its paired mutation — proves M3 did not turn any rule into a bluff), every propagation gate (literal anchors intact), a strong-model reasoning-path probe (proves M2/M5 tiered nothing that decides), and a cache-warm proof (`cache_read_input_tokens > 0`). Cost reduction with quality/safety regression is a §11.4 FAIL, not a win.

No measure may break or degrade any existing mechanism — and the rule is structured so none can: caching is transparent (M1); tiering and low-effort are confined to mechanical non-judgment work (M2/M5); indexing preserves the full rules by reference with literal anchors so the propagation gates pass and the canonical text stays gate-scanned (M3); retrieval uses already-trusted read-only infrastructure (M4). The §11.4 anti-bluff covenant family, §11.4.50 deterministic consistency, §11.4.125 code-review, §11.4.40 full-suite retest, and §12.6 memory ceiling are all unconditionally preserved.

Composes with §11.4.5 / §11.4.6 / §11.4.20 / §11.4.40 / §11.4.50 / §11.4.58 / §11.4.69 / §11.4.70 / §11.4.78 / §11.4.79 / §11.4.80 / §11.4.103 / §11.4.106 / §11.4.123 / §11.4.125 / §11.4.128 / §12.6 / §1.1. Classification: universal (§11.4.17) — the consuming project supplies its model IDs, its mechanical-subagent allowlist, its governance files, its structural-index tool, and its frozen workload per §11.4.35. Propagation gate `CM-COVENANT-114-141-PROPAGATION` (literal `11.4.141` across the consumer fleet) + recommended gate `CM-TOKEN-EFFICIENCY` (the measured BEFORE/AFTER harness exists + produced a verdict ≥ target-or-cited-floor + the non-regression sweep is green + cache-warm proof present) + paired §1.1 meta-test mutation (inject a per-turn volatile token ahead of the governance cache breakpoint → `cache_read_input_tokens` collapses → measured reduction falls below the bar → gate FAILs; restore → PASSes — proves the harness measures real caching, not a hardcoded green). Gate-code = separate work item.

Canonical authority: constitution submodule `Constitution.md` §11.4.141. Research + design + measurement: `docs/research/token_efficiency/{RESEARCH,DESIGN,MEASUREMENT,TEST_PLAN}.md` .

Non-compliance is a release blocker regardless of context. No escape hatch — no `--skip-token-efficiency`, `--no-cache-governance`, `--assert-reduction-without-measuring`, `--tier-down-reasoning`, `--inline-all-governance`, `--tiktoken-estimate-OK` flag exists.

(B) Consumer-side mirror block — for every consumer CLAUDE.md / AGENTS.md / QWEN.md

§11.4.141 — Token-efficiency mandate (research-derived + operator mandate, 2026-06-09). Every project worked on by AI coding agents MUST cut token spend (input AND output) toward **30–40% of current (a 60–70% reduction)** WITHOUT degrading quality/performance/safety or breaking any existing mechanism, via a composable, safety-ranked measure set: (1) **prompt-cache the static governance prefix** — the always-loaded governance forms a byte-stable cache-breakpointed prefix with no volatile bytes ahead of it; cache reads cost ~0.1× base input (the dominant cost driver — measured ~170K tokens of governance re-sent every turn, externally corroborated by Claude Code issue #24147); caching is transparent so it removes no rule, weakens no gate, changes no verdict — only billing (PRIMARY, biggest + safest lever); (2) **subagent model-tiering + output-to-file** — mechanical non-judgment work (search/grep/status/doc-export/read-only probes) to a Haiku-class model, the strong model RESERVED for all reasoning/verdicts/fix-design (§11.4.102)/code-review (§11.4.125)/demotion (§11.4.7), large output persisted to a file not an inline 350–520K-token transcript; the cheap model never emits a PASS so §11.4.50 + anti-bluff are untouched; (3) **thin always-loaded INDEX + on-demand detail** — concise index (one line per fix/anchor, EACH carrying the literal `11.4.N` token so propagation gates pass) with the canonical full text kept gate-scanned in `constitution/Constitution.md` and reachable in one hop — a de-duplication realising §11.4.35, never a deletion; (4) **CodeGraph/retrieval-first over full-file loading** (§11.4.78/§11.4.79); (5) **output-token reduction** — terse status + `effort:"low"` on the mechanical allowlist only; (6) **tool-call batching + no re-reads**; (7) **compaction/context-editing for long sessions**. **Mandatory measured proof:** a token-accounting harness measures tokens-per-development-cycle BEFORE vs AFTER on a frozen deterministic workload from the authoritative

`usage` object (input/cache_read/cache_creation/output split; NEVER `tiktoken`, NEVER the client-side cost estimate), reproduced N times (§11.4.50), pass = AFTER ≤ 40% of BEFORE OR the measured best-safe reduction with a cited cold-cache reason; the AFTER run MUST show ZERO regression on the pre-build sweep + meta-test mutation sweep + propagation gates + a strong-model reasoning probe + a cache-warm proof (`cache_read_input_tokens > 0`) — cost reduction with quality regression is a §11.4 FAIL. The headline number is the *measured* reduction, never the design estimate (§11.4.6/§11.4.123). No measure may break/degrade any existing mechanism, and the rule is structured so none can. Composes §11.4.5/.6/.20/.40/.50/.58/.69/.70/.78/.79/.80/.103/.106/.123/.125/.128/§12.6/§1.1. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-141-PROPAGATION` (literal `11.4.141`) + recommended gate `CM-TOKEN-EFFICIENCY` + paired §1.1 mutation (inject a pre-breakpoint volatile token → cache collapses → measured reduction falls below bar → gate FAILs). **Canonical authority:** constitution submodule `Constitution.md` §11.4.141. **Project instantiation (§11.4.35):** [ATMOSphere consumer fills in: confirm the Claude Code governance-prefix cache is warm + free of pre-breakpoint volatiles; tier the §11.4.96-SAFE mechanical subagents (search/grep/status/doc-export) to Haiku with output-to-file under `qa-results/` ; replace the consumer `CLAUDE.md` 112-row Applied-Fixes inline table + verbatim anchor restatements with a literal-anchor index pointing at `constitution/Constitution.md` ; keep CodeGraph/lumen-first navigation; run the harness BEFORE/AFTER to prove the measured reduction.] Non-compliance is a release blocker. No escape hatch — no `--skip-token-efficiency` , `--no-cache-governance` , `--assert-reduction-without-measuring` , `--tier-down-reasoning` , `--inline-all-governance` , `--tiktoken-estimate-OK` flag.